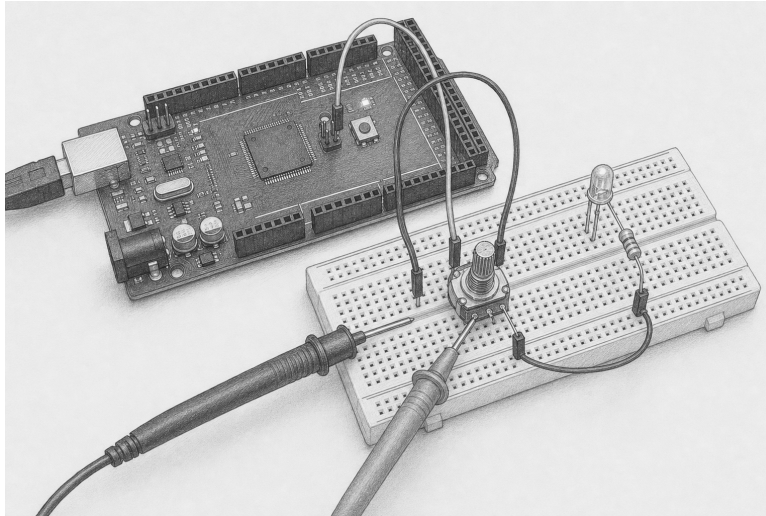


# Make every transformation visible

## Lesson 008: calibrate and stabilize a sampled signal

*Predict → sample → map → average → hold → observe*



**Orientation plate.** Use the pencil view to identify the three observable places: the knob, **TP-A0**, and the brightness LED. It is not a pin map. The exact net list and schematic below are authoritative.

|                      |                                                                                                                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Question</b>      | Can explicit, replayable transformations reduce visible jitter without hiding what the circuit measured?                                                                                   |
| <b>Time</b>          | 75–100 minutes                                                                                                                                                                             |
| <b>Prerequisites</b> | Lesson 007; integer ranges; explicit status handling                                                                                                                                       |
| <b>You need</b>      | Mega 2560, USB cable, breadboard, 10 kΩ linear potentiometer, LED, resistor of at least 220 Ω, jumpers, multimeter, and logic analyzer or oscilloscope with a 10 kΩ/10 kΩ midpoint fixture |
| <b>Evidence</b>      | TP-A0 voltage, raw and transformed worksheet, D6 duty/brightness, D13 acquisition witness, safe-state record                                                                               |
| <b>Safety class</b>  | E1: USB-powered Mega and passive low-current parts only                                                                                                                                    |
| <b>Status</b>        | Host verified; physical acceptance pending                                                                                                                                                 |

### Safety boundary

Remove USB before wiring or moving a probe connection. Use a 10 kΩ potentiometer between Mega 5 V and GND; connect only its wiper to A0. Drive the external LED from D6 through its own resistor of at least 220 Ω. All grounds are common.

Use a regulated 5 V USB source rated for at least 500 mA and record its label rating. Identify the LED color, package, polarity, and datasheet or measured forward voltage  $V_f$ . Before power, calculate

$$I_{\text{LED}} = \frac{V_{\text{source}} - V_f}{R}.$$

Proceed only when the worst-case result is at most 15 mA; otherwise increase the resistor. This limit is a lesson go/no-go rule, not a claim that the full source rating may flow through an I/O pin.

Filtering is interpretation, not electrical protection. It cannot make an over-voltage safe, prove a wire is sound, or turn a floating input into valid evidence. A rail endpoint can be a real knob position or a fault. A disconnected A0 can wander through plausible values.

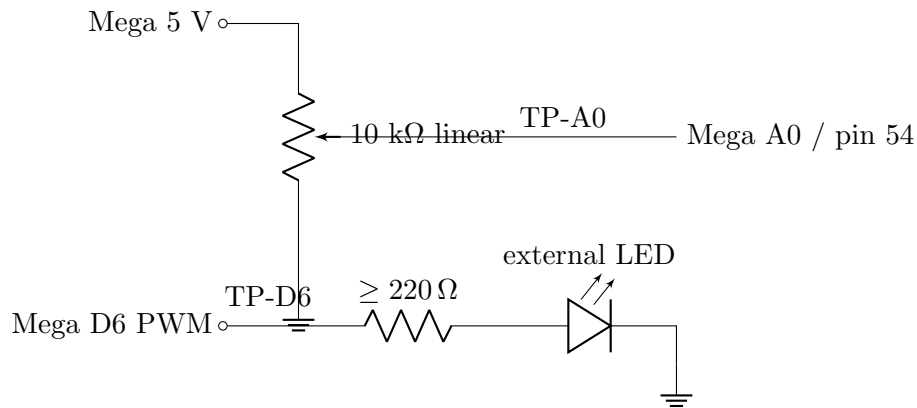
**Stop** for heat, odor, smoke, USB disconnects, a voltage outside 0–5 V, an unidentified potentiometer terminal, or an unknown resistor. Disconnect USB upstream; do not move powered breadboard wiring.

**Check — before wiring**

☐ USB removed    ☐ pot resistance and terminals identified    ☐ LED resistor measured    ☐ common ground traced

## 1. Reuse a controlled source

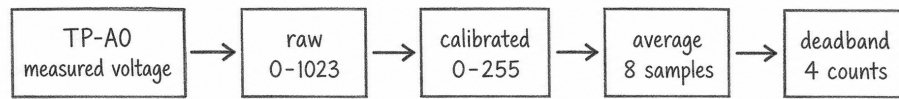
A controlled potentiometer is deliberately retained from Lesson 007. It lets you hold, step, and repeat the input before Lesson 009 adds a photoresistor’s environmental variability.



| From                   | Through                 | To               | Purpose               |
|------------------------|-------------------------|------------------|-----------------------|
| Mega 5 V               | pot outer high terminal | —                | known source          |
| pot center wiper       | TP-A0                   | Mega A0 / pin 54 | sampled position      |
| pot outer low terminal | —                       | Mega GND         | divider return        |
| Mega D6                | resistor                | LED anode        | current-limited drive |
| LED cathode            | jumper                  | Mega GND         | brightness return     |
| Mega D13               | built-in LED            | —                | acquisition witness   |

## 2. Predict the evidence chain

The circuit and program make five different claims. Do not collapse them into “the LED works.”



D13's 250 ms pulse supports the separate claim that all three hardware owners initialized. A 750 ms dark separator follows before sampling begins. D6 brightness and its PWM waveform expose the final held value. TP-A0 remains available to challenge every software interpretation.

**Predict.** Before power, write what should happen after a quick knob step and what should happen while the knob is held still:

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### 3. Calibrate observations, not ideal endpoints

The canonical example maps observed readings 80 through 940 onto duty 0 through 255 and clamps values beyond those observations. These numbers are a repeatable exercise, not universal facts about your potentiometer.

For an increasing calibration,

$$y = y_{\min} + \frac{(x - x_{\min})(y_{\max} - y_{\min})}{x_{\max} - x_{\min}},$$

with integer rounding. `LinearCalibration` validates the interval and returns `Result<uint16_t>` because an invalid configuration has no meaningful mapped result.

```

const adk::LinearCalibrationConfig calibrationConfig{
    80, 940, 0, 255, true};

adk::LinearCalibration calibration (calibrationConfig);

const adk::Result<uint16_t> mapped = calibration.map (raw);
if (!mapped.ok ())
{
    stopSafely ();
}
  
```

| Position      | TP-A0 | Raw observation | Mapped prediction | Mapped result |
|---------------|-------|-----------------|-------------------|---------------|
| low stop      |       |                 | 0 after clamp     |               |
| quarter       |       |                 |                   |               |
| middle        |       |                 |                   |               |
| three-quarter |       |                 |                   |               |
| high stop     |       |                 | 255 after clamp   |               |

Measure Mega 5 V and TP-A0 with power on only after placing probes while unpowered. The ideal relation is

$$\text{raw} \approx \frac{V_{\text{TP-A0}}}{V_{\text{ref}}} \times 1023.$$

Use measured  $V_{\text{ref}}$ , not an assumed perfect 5.000 V.

## 4. Advance state only from supplied samples

**MovingAverage** owns a fixed array of at most 32 readings. The example uses eight. It has no clock and takes no hidden sample: each `addSample()` advances it exactly once. Early results average the samples received so far.

**Deadband** publishes the first sample, then retains that value until a new sample differs by at least its width. It prevents small visible output changes; it does not erase raw evidence.

```

adk::MovingAverage average      (8);
adk::Deadband      brightnessHold (4);

const adk::Result<uint16_t> smoothed =
    average.addSample (mapped.value ());
const uint16_t stable =
    brightnessHold.addSample (smoothed.value ());

```

Order matters. Mapping first gives the deadband a meaning in output duty counts. Averaging before the deadband lets a real sustained change accumulate. Reversing these objects would describe a different instrument.

## 5. Replay a deterministic sample stream

Calculate the eight-sample, rounded moving average, then apply a deadband of four. The initial published value is 100.

| Step | Supplied sample | Window contents                   | Rounded average | Held output |
|------|-----------------|-----------------------------------|-----------------|-------------|
| 1    | 100             | 100                               | 100             | 100         |
| 2    | 102             | 100, 102                          |                 |             |
| 3    | 101             | 100, 102, 101                     |                 |             |
| 4    | 104             | 100, 102, 101, 104                |                 |             |
| 5    | 112             | 100, 102, 101, 104, 112           |                 |             |
| 6    | 116             | 100, 102, 101, 104, 112, 116      |                 |             |
| 7    | 120             | 100, 102, 101, 104, 112, 116, 120 |                 |             |
| 8    | 124             | all eight                         |                 |             |

Replay the same values twice. The result must match at every step. Change only one sample and identify the first output that can change. This is the central testability property: the state is a function of configuration and supplied sample history, not processor speed or wall-clock timing.

## 6. Read the narrative run loop

The canonical sketch introduces owners before transformations. Setup validates the software configuration, acquires D13, A0, and D6, then pulses D13 for 250 ms and leaves a 750 ms dark separator. The loop reads as observe, decide, actuate:

```
potentiometer.update ();

adk::PwmOutput::Duty brightness = 0;
if (!chooseBrightness (brightness))
{
    stopSafely ();
    return;
}

if (!brightnessLed.write (brightness).ok ())
{
    stopSafely ();
}
```

The 20 ms sampling schedule belongs to the application. Calibration, averaging, and deadband remain useful in host tests without Arduino time. Each observation is also emitted as a four-column CSV row: raw, calibrated, average, then held. That optional textual trace makes every software stage recordable, while TP-A0 and D6 remain the required circuit evidence. After 120 seconds the sketch shuts down without user input, leaves a 250 ms measurement window, and halts. Shutdown turns off D6, makes A0 and D6 high impedance, releases all claims, and turns off D13. Destruction repeats that cleanup safely. Reset starts a new acquisition pulse and timed run.

## 7. Prove the transforms on the host

Write deterministic tests before interpreting the breadboard:

1. invalid and reversed calibration intervals;

2. exact endpoints, midpoint rounding, clamping, and unclamped rejection;
3. moving-average windows 0, 1, 8, 32, and 33;
4. partial fill, full fill, replacement, reset, and 16-bit extremes;
5. deadband equality, just inside, just outside, rising, falling, reset;
6. two identical sample streams with identical result/status traces;
7. A0 claim conflict and D6 timer conflict before any unintended write;
8. repeated shutdown and destruction from an active circuit.

```
make host-test
make quality-fast
make arduino-Lesson008SampledSignal
```

Compilation and simulation do not establish physical acceptance.

## 8. Upload and observe without depending on Serial

Record the exact specimen before upload: Mega board marking, board serial or lab asset ID, USB supply identity, potentiometer marking and measured end-to-end resistance, LED color/package, measured series-resistor value, and jumper/breadboard identifier. Record the meter and analyzer manufacturer, model, asset or serial ID, probe or lead identity, and last-calibration date. Identify the board and port, compile the one canonical sketch, upload it, and optionally capture the CSV transformation trace:

```
make boards
make arduino-Lesson008SampledSignal
make upload EXAMPLE=Lesson008SampledSignal PORT=/dev/ttyACM0
make monitor PORT=/dev/ttyACM0 BAUD=115200
```

The required evidence is circuit-native:

1. D13 emits one 250 ms pulse, then remains dark for the 750 ms separator: resource-acquisition witness. Measure its duration.
2. TP-A0 voltage follows the knob: primary input evidence.
3. D6 duty and LED brightness follow a sustained change: output evidence.
4. Small TP-A0 movement may leave D6 unchanged: deadband evidence.
5. A fast step reaches D6 gradually: moving-average evidence.
6. At 120 seconds, D6 and D13 become inactive without a button or host command. During the following 250 ms window, the analyzer records the transition and the midpoint fixture checks released pins.

| Experiment         | Prediction                                      | Observation | Interpretation |
|--------------------|-------------------------------------------------|-------------|----------------|
| hold at middle     | TP-A0 and D6 settle                             |             |                |
| small nudge        | TP-A0 moves; D6 may hold                        |             |                |
| large step         | TP-A0 jumps; D6 ramps                           |             |                |
| return step        | same direction-independent rules                |             |                |
| automatic shutdown | at 120 s, D13/D6 inactive; pins released        |             |                |
| power removal      | after shutdown, 5 V and test points de-energize |             |                |

**Released-pin fixture.** Remove USB, connect two measured 10 k $\Omega$  resistors in series from 5 V to GND, and connect their midpoint to only the pin under test. Reapply USB and run the experiment. During the 250 ms shutdown window, measure approximately half the supply at D6; then repeat a separate run for A0. Never attach, remove, or move this fixture while powered, never test two pins at once, and never open the ground connection while powered. LED darkness alone is not high-impedance evidence.

**Reset and acquisition rollback.** Press reset once after a completed run. Record a new 250 ms acquisition pulse and confirm the CSV column order and response repeat from an identical knob position. Then compile a separate fault-injection artifact:

```
arduino-cli compile --fqbn arduino:avr:mega --library . \
  --build-property \
  compiler.cpp.extra_flags=-DADK_LESSON008_INJECT_D6_CONFLICT \
  examples/Lesson008SampledSignal
```

Upload that artifact only for the rollback run. It deterministically claims D6 before normal acquisition. The subsequent D6 acquisition must fail; the sketch must emit no acquisition pulse and must roll back the already acquired A0 and D13 owners before releasing the injected D6 claim. In separate powered runs, with the midpoint fixture installed while USB is absent, record released-state evidence for A0 and D13. Remove USB after each run. Recompile and upload the canonical artifact before continuing.

**Configuration restoration.** To demonstrate return to the prerequisite configuration, remove USB, restore the unmodified Lesson 007 potentiometer/LED wiring, inspect it against that lesson's net list, upload its canonical sketch, and record its acquisition and finite safe-state evidence. This is configuration restoration, not the acquisition-failure rollback test. If upload, identity, timing, or fixture checks disagree, remove USB and return to the last inspected unpowered wiring state; do not improvise around a powered fault.

**Physical power removal.** After the canonical automatic-shutdown run, record the precondition: D13 and D6 inactive and the analyzer capture saved. Unplug USB at the upstream source without touching breadboard wiring. Record the observation: Mega power LED, D13, and external LED are dark and TP-A0, TP-D6, and 5 V measure de-energized. Interpret darkness plus voltage measurements as removal of source power; do not interpret it as pin-release evidence, which belongs to the powered midpoint-fixture checks.

## 9. Diagnose by stage

| Observation                    | Next measurement                                                | What it can establish                                              |
|--------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------|
| D13 stays off                  | inspect status path and unpowered wiring                        | acquisition failed; not which owner failed without recorded status |
| D13 on, TP-A0 fixed            | measure wiper and both pot ends to GND                          | distinguishes source wiring from later transforms                  |
| TP-A0 moves, D6 fixed          | inspect expected mapped/average/deadband worksheet; probe TP-D6 | separates decision from LED wiring                                 |
| D6 waveform changes, LED fixed | remove power; inspect polarity and resistor                     | the command reached the pin but the load path is suspect           |
| rail reading                   | inspect all three pot terminals unpowered                       | may be endpoint or fault; filtering is not evidence either way     |
| plausible wandering input      | power off; check A0 continuity                                  | a disconnected input can float plausibly                           |

## 10. Extend the design

1. Reduce the average window from eight to two. Predict response time and visible jitter before running it.
2. Increase the deadband from four to sixteen. Which deliberate knob movements disappear?
3. Swap average and deadband in a host-only experiment. Explain why it is a different policy.
4. Record ten raw endpoint readings, choose observed calibration bounds, and justify the margin retained at each end.
5. Replace the potentiometer samples with a recorded photoresistor trace on the host. Do not change hardware yet. Which parameters need reconsideration for Lesson 009?



## Bench acceptance record

Board and supply

Board marking, serial / asset ID

USB supply and cable identity

Pot, LED, resistor, breadboard / jumpers

Meter make, model, ID, calibration date

Analyzer / scope, probe, ID, calibration date

Measured 5 V reference

Source rating; LED identity and  $V_f$

Measured pot and LED resistor

Calculated worst-case LED current /  $\leq 15$  mA go/no-go

Observed calibration bounds

D13 acquisition observation

CSV raw / calibrated / average / held file

TP-A0 observations

TP-D6 / brightness observations

120 s deadline and 250 ms window

D6 and A0 midpoint-fixture evidence

Reset repeat and injected D6 rollback evidence

Lesson 007 configuration-restoration evidence

Power removal P / O / I

Deviations, reviewer, date

Check — exit

☐ source circuit checked unpowered    ☐ every transformation predicted    ☐ deterministic replay passed    ☐ non-Serial evidence recorded    ☐ resource and safe-state evidence kept separate

**Companion material.** Use the HTML lesson for current API signatures, canonical source links, errata, and searchable procedures. Use this PDF for predictions, hand calculations, stage-by-stage diagnosis, and the physical acceptance record. Until that record is completed by a person at the named bench, this lesson remains host-verified.

Arduino Mega 2560 documentation    Microchip ATmega2560 datasheet